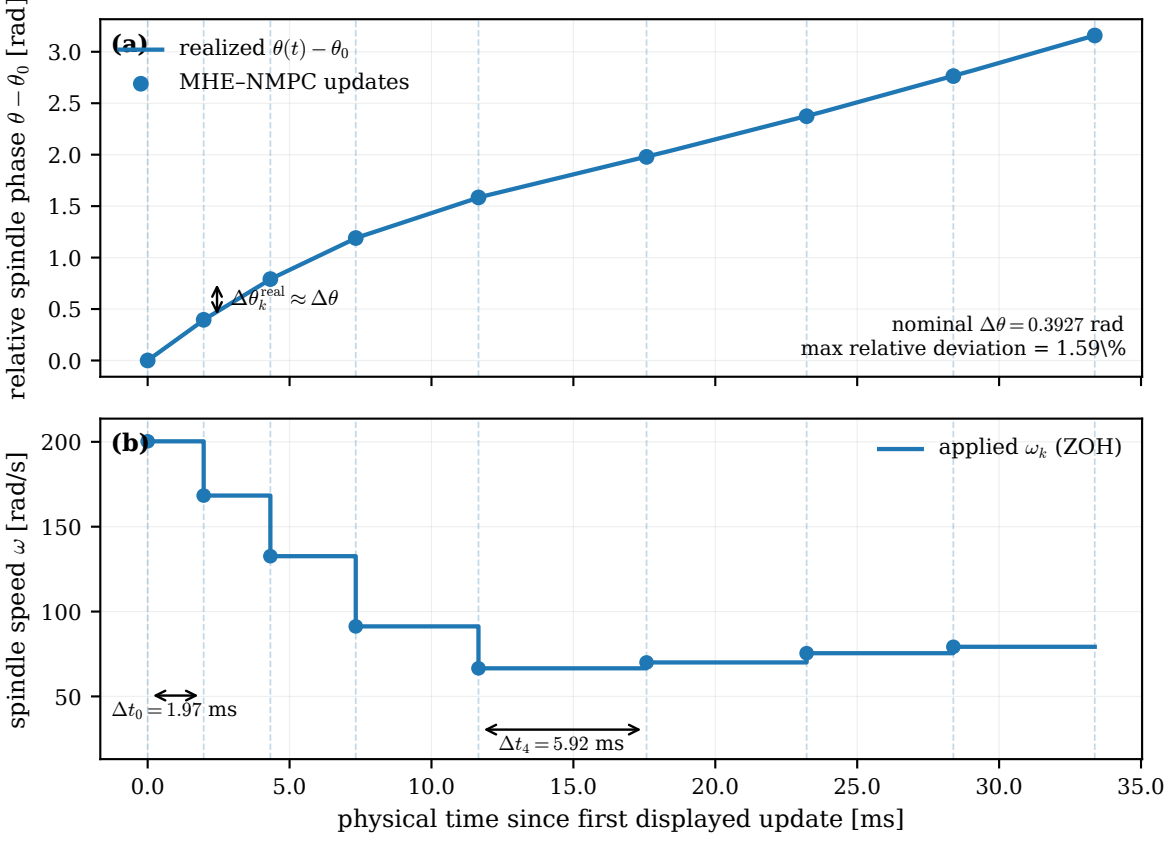
(a)
realized $\theta(t) - \theta_0$
MHE–NMPC updates
relative spindle phase $\theta - \theta_0$ [rad]
$\Delta\theta_k^{\text{real}} \approx \Delta\theta$
nominal $\Delta\theta = 0.3927$ rad
max relative deviation = 1.59\%
(b)
applied ω_k (ZOH)
spindle speed ω [rad/s]
$\Delta t_0 = 1.97$ ms
$\Delta t_4 = 5.92$ ms
physical time since first displayed update [ms]